Nilai UNIVERSITY

BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION
FEBRUARY 2025

Course:	EC3130 (Introduction to Data Communication and Networking)	Time:	8:00 am – 11:00am (3 hours)
Lecturer:	Mr. SEOW Soon Loong	Date:	22 May 2025

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **3** printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Compare and contrast the following network topologies: Star, Ring, and Mesh. For each topology, provide: **(18 MARKS)**
   - A brief description.
   - At least two advantages and two disadvantages.

   b. In the context of network protocol models, match the following layers of the OSI model with their corresponding layers in the TCP/IP model: **(7 MARKS)**
   - Application Layer (TCP/IP Model)
   - Transport Layer (TCP/IP Model)
   - Internet Layer (TCP/IP Model)
   - Network Access Layer (TCP/IP Model)

   Provide the corresponding OSI layers for each of the TCP/IP model layers listed above.

2. a. Compare and contrast between optical fiber and copper cabling. **(10 MARKS)**

   b. Explain the Carrier-Sense Multiple Access with Collision Avoidance (CSMA/CA) protocol. Describe the steps a station follows when it wants to transmit a frame. **(15 MARKS)**

3. a. A signal has a period of 200 Milliseconds (ms). You are tasked with designing a system that requires the signal's frequency to be calculated in kilohertz. **(3 MARKS)**

   b. A digitized audio channel is created by digitizing a 15-MHz bandwidth analog audio signal. To sample the signal, we must sample at twice the highest frequency (two samples per hertz). Assume each sample requires 12 bits. What is the required bit rate for this digitized audio signal? **(4 MARKS)**

   c. Given a binary data stream "011001", describe how the Polar Return-to-Zero (RZ) encoding scheme would represent this data. Illustrate you answer with a diagram. **(6 MARKS)**

   d. Compare and contrast the Manchester and Differential Manchester encoding schemes in terms of signal transitions. Provide an example of how each scheme would behave with a data sequence of "100110". Illustrate you answer with diagrams. **(12 MARKS)**

4. a. You are assigned to design an IPv4 network for a company's IT infrastructure using Classless Addressing. The company has been allocated the Class C IP address 192.168.10.0/24 and requires network segmentation for efficient IP allocation. **(14 MARKS)**

The network consists of the following components:

- **FIVE (5)** Local Area Networks (LANs) for different departments: IT Support (20 hosts), Finance (10 hosts), Sales (50 hosts), Customer Service (25 hosts), Logistics (12 hosts)
- **TWO (2)** Wide Area Networks (WANs), WLinkA and WLinkB, for connecting remote sites.

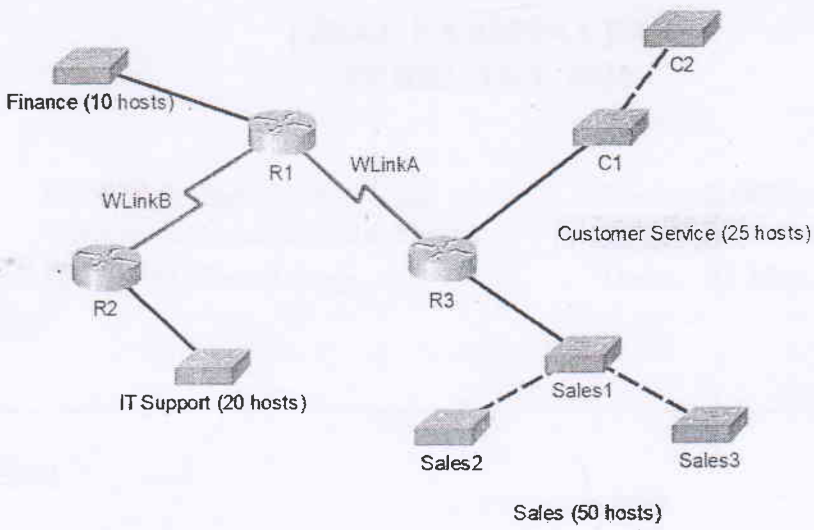

Figure 4.1

For each subnet, provide the following details in the given table format:

Subnet Name	Subnet Address	First Usable IP	Last Usable IP	Broadcast Address	Subnet Mask	Number of Addresses
Sales	192.168.10.0/26	192.168.10.1				
Customer Service	192.168.10.64/27	192.168.10.65				
IT Support	192.168.10.96/27	192.168.10.97				
Finance	192.168.10.128/28	192.168.10.129				
Logistics	192.168.10.144/28	192.168.10.145				
WLinkA	192.168.10.160/30	192.168.10.161				
WLinkB	192.168.10.164/30	192.168.10.165				

Table 4.2

b. Provide a step-by-step example of shortening the following IPv6 address using the rules for IPv6 address compression:

i. 2001:0db8:0:0042:0000:8a2e:0370:7334 **(2 MARKS)**

ii. fe80:0000:0000:0000:0202:b3ff:fe1e:8329 **(2 MARKS)**

c. Describe the principles of channelization in Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and Code Division Multiple Access (CDMA). **(7 MARKS)**

-END OF QUESTION PAPER-